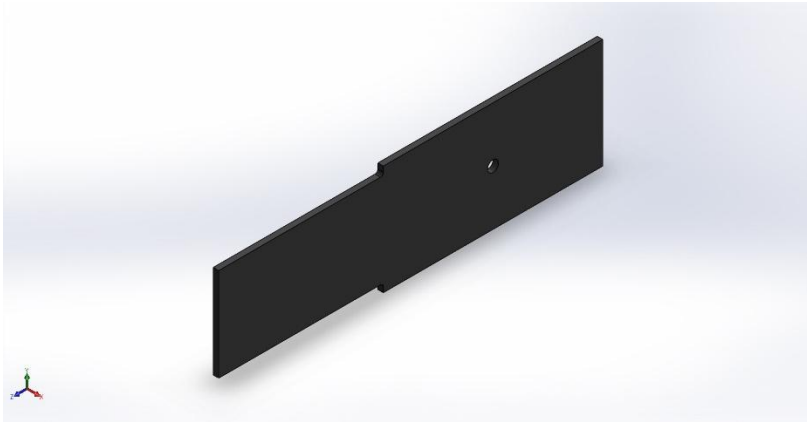




Simulation of bar

Date: Saturday, January 14, 2023

Designer: Solidworks

Study name: Bar 1

Analysis type: Static

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Description

Section_Properties_Summary	
Property	Value/Description
Bar Thickness	2 mm
Bar Width	40 mm
Bar Length	34 mm
Hole Diameter (Case 1)	4 mm
Hole Diameter (Case 2)	8 mm
Fillet Radius	1 mm
Material	Heat-treated and quenched steel (brittle)
Young's Modulus (E)	~200 GPa
Axial Load	10 kN
Boundary Conditions	Fixed at one end, axial load at the other
Nominal Stress (σ_0) for 4 mm Hole	139 MPa
Stress Concentration Factor (K_t) for 4 mm Hole	2.7
Maximum Stress (σ_{max}) for 4 mm Hole	380 MPa
Assumptions	Linear elastic, homogeneous, isotropic

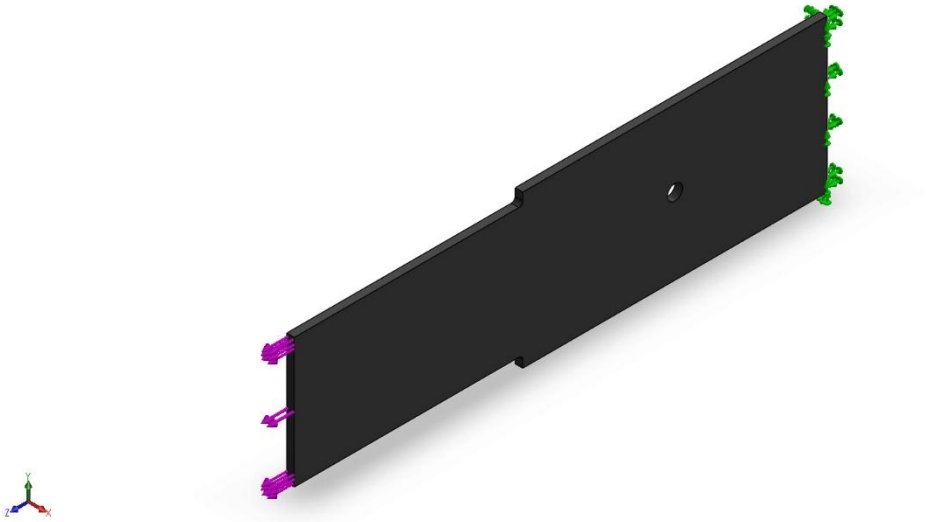


SOLIDWORKS

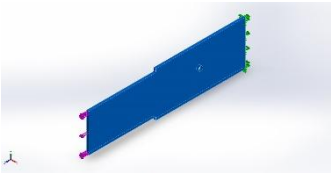
Analyzed with SOLIDWORKS Simulation

Assumptions

Model Information



Model name: bar
Current Configuration: Default

Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Cut-Extrude1 	Solid Body	Mass:0.0752812 kg Volume:1.04557e-05 m^3 Density:7,200 kg/m^3 Weight:0.737756 N	E:\L-3 T-2\Assignments\Part 3\bar.SLDPRT Jan 13 23:52:45 2023



Study Properties

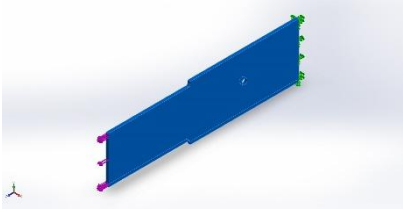
Study name	Bar 1
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (E:\L-3 T-2\Assignments\Part 3)

Units

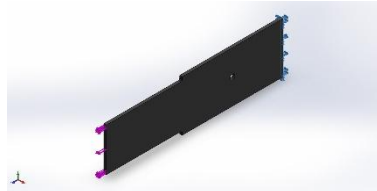
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²



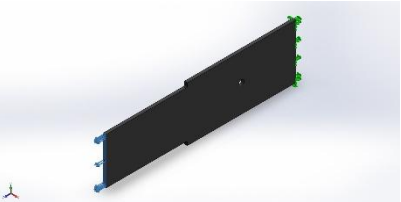
Material Properties

Model Reference	Properties	Components
	Name: Gray Cast Iron (SN) Model type: Linear Elastic Isotropic Default failure criterion: Mohr-Coulomb Stress Tensile strength: 1.51658e+08 N/m ² Compressive strength: 5.72165e+08 N/m ² Elastic modulus: 6.61781e+10 N/m ² Poisson's ratio: 0.27 Mass density: 7,200 kg/m ³ Shear modulus: 5e+10 N/m ² Thermal expansion coefficient: 1.2e-05 /Kelvin	SolidBody 1(Cut-Extrude1)(bar)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-0.133416	-0.0448534	-10,001.2	10,001.2
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s) Type: Apply normal force Value: -10,000 N



Connector Definitions

No Data

Contact Information

No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	1.09376 mm
Minimum element size	0.364582 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	104927
Total Elements	62425
Maximum Aspect Ratio	3.351
% of elements with Aspect Ratio < 3	100
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:11
Computer name:	HELAL



Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.133416	-0.0448534	-10,001.2	10,001.2

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.618066	-2.32296	4.44743	5.05546

Free body moments

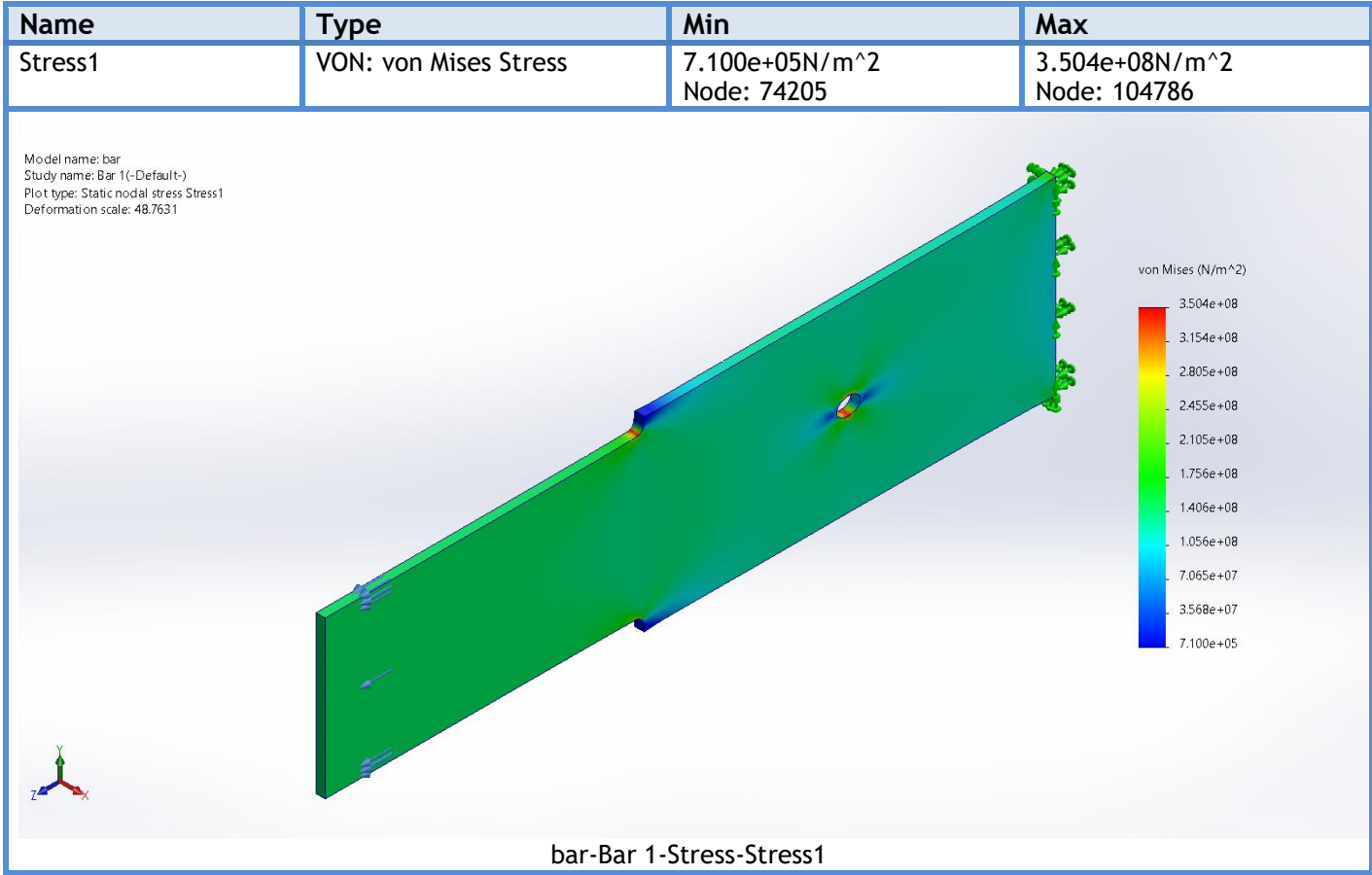
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data



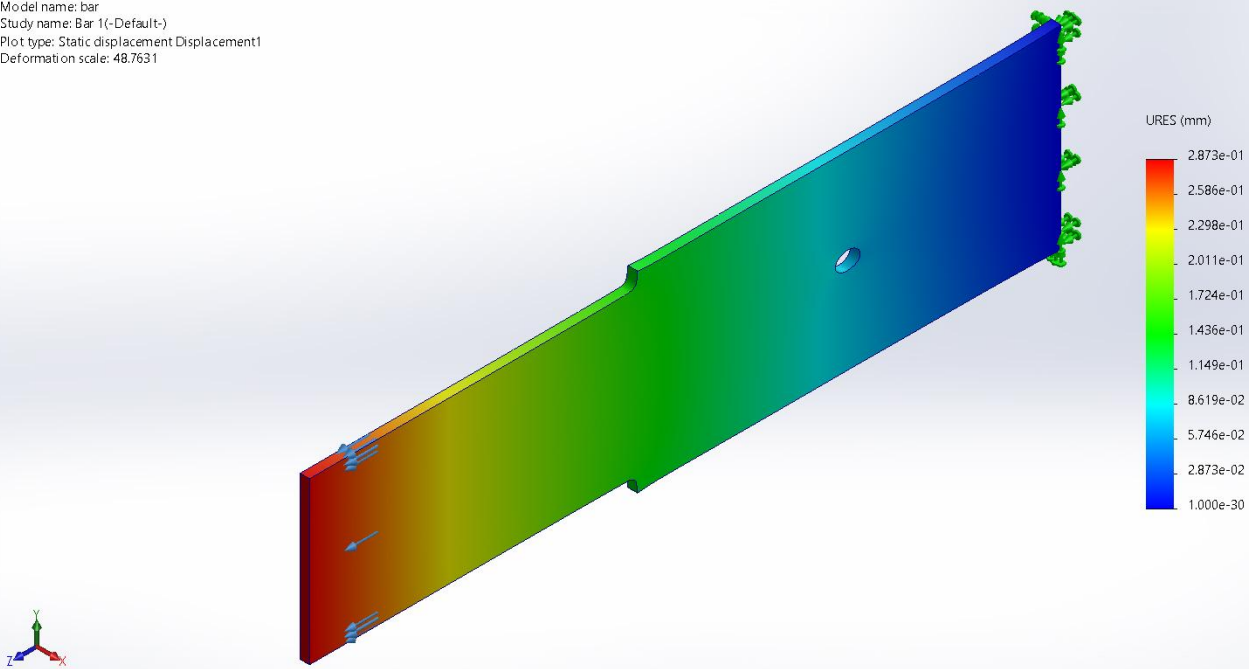
Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 11	2.873e-01mm Node: 23



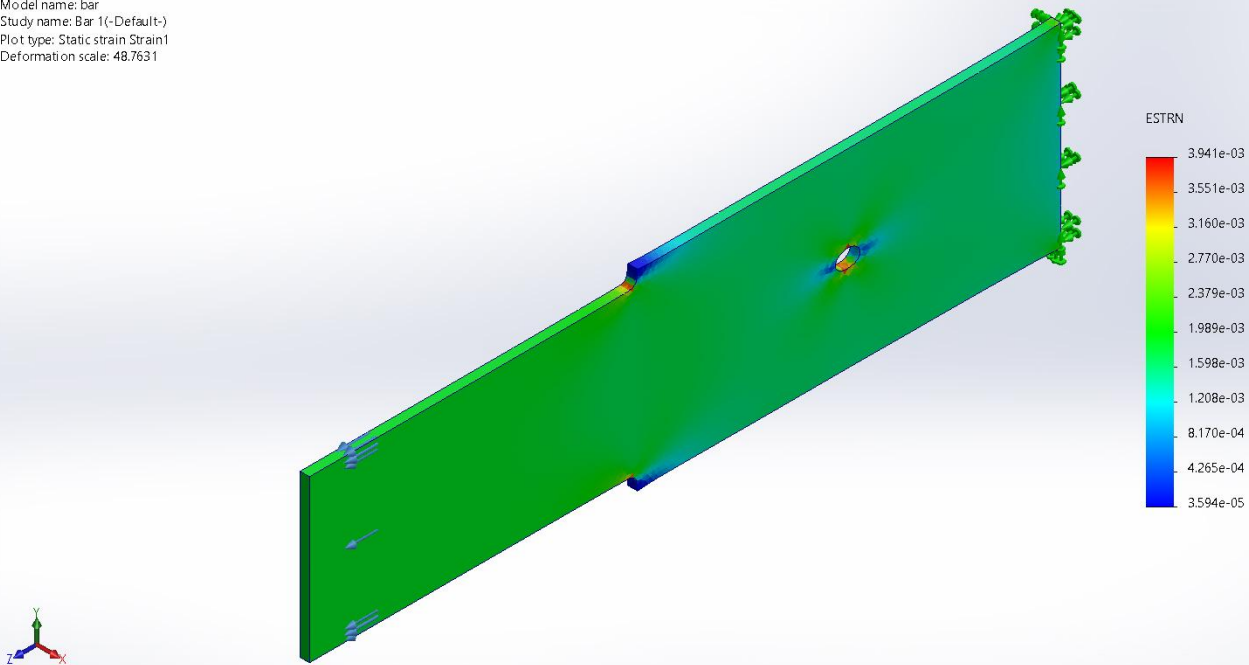
Model name: bar
 Study name: Bar 1(-Default-)
 Plot type: Static displacement Displacement1
 Deformation scale: 48.7631



bar-Bar 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	3.594e-05 Element: 23073	3.941e-03 Element: 39473

Model name: bar
 Study name: Bar 1(-Default-)
 Plot type: Static strain Strain1
 Deformation scale: 48.7631



bar-Bar 1-Strain-Strain1



Conclusion

